

Comparative Study of Convolutional Neural Networks for Leaf Classification in Ayurveda

Sweetey Kunjachan¹, Remya R¹, Santhos Kumar¹, Asish G R², Sheela K², and Kala S¹

¹ Department of ECE, Indian Institute of Information Technology (IIIT) Kottayam, Kerala-686635, India

² Vaidyaratnam Ayurveda Research Institute (VARI), Thaikkattussery, Thrissur, Kerala, India

*Email: {sweetygisat.22phd12003, rr.madathil.22phd12001, santhos, kala}@iiitkottayam.ac.in

*Email: {rd.asish, rd.head}@vaidyaratnam.com

Abstract

Ayurveda, the traditional system of Indian medicine and healthcare has proven itself over the past centuries. It has grown in acceptance and repute in the eyes of Indians and is also fast spreading to all corners of the world. Herbs are the main essence and are at the core of most ayurvedic medicines. A reduction in the number of experts in the field of identification and sourcing of herbs reduces the quality of medicines. Recognition of herbs through automatic classification of leaves is a remedy for this problem. A vast number of deep learning models for classification of images have emerged in the past decade. This paper makes a comparison between various state-of-the-art convolutional neural networks for classification of leaves. We evaluate architectures including VGG-16, ResNet-50, DenseNet-121, MobileNetV2, EfficientB0 and EfficientV2M. This experiment has been done with a data of 1835 images of leaves taken from 30 different species. ResNet-50, VGG-16, EfficientV2M returned the best result with an accuracy of 98% and EfficientV2M attained the highest accuracy with less number of epochs.

1 Introduction

Ayurveda is the gleaming crown jewel amongst India's various contributions to the betterment of mankind [1]. Ayurveda believes that health is a mirror of harmony between nature and mankind. Ayurveda focuses mainly on root causes of diseases and tries to rectify them through natural remedies. Herbs play a vital role in the preparation of ayurvedic medicines. According to FFPRI (Forestry and Forest Product Research Institute) in Japan, forest area in relation to population has shown a decline of 60% over the last six decades [2]. As a result of decreasing forest area, the availability of medicinal plants has become scarce. General awareness and knowledge of medicinal plants is also decreasing due to multiple socio-economic reasons. A huge cache of knowledge about medicinal plants resides with villagers and tribals living predominantly in the interiors of forests. This knowledge, accumulated and aggregated over generations remains unknown and untapped to the modern community living far off in cities. Thus, the need of the hour is to preserve this knowledge in digital form.

Identification of plants is manually done by observing the morphological characteristics such as root, stem, fruits, leaves etc. Among them, leaves are found above the ground and will be present over the life of a plant while fruits and flowers remain only for a relatively small period of time [3]. Hence researchers mainly focus on classification of leaves for identification of herbs.

Artificial Intelligence (AI) can be tasked with computer aided plant disease classification, herb recognition etc due to its multi faceted, varied and immense potential [4]. Ad-

vancements in deep learning models are very crucial for these studies. There have been many inspirational quantum leaps in image classification through deep convolutional neural network [5]. The pretrained models coming under this category help developers to build AI models without being explicitly built from scratch [6]. Transfer learning solves a problem with experience gained from previously acquired knowledge. It may be used as is or further fine-tuned to fit an application's specific needs [7]. A model that has already been pre-trained on a large dataset saves training time period and reduces computational cost whilst also delivering better accuracy with a smaller number of image dataset. This study is an attempt on an empirical analysis of pretrained models commonly entrusted with the task of leaf classification. The architectures used are VGG-16, ResNet-50, DenseNet-121, MobileNetV2, EfficientB0 and EfficientV2M.

The rest of the paper is divided into four sections. Section 2 takes account of the related works in the field of leaf classification. Section 3 describes existing state-of-the-art techniques required to perform the experiment. Section 4 details the experimental setup as well as results and the final Section 5 concludes the paper.

2 Related Works

Convolutional Neural Networks (CNN) have garnered the faith of industries and researchers by heralding deep learning techniques to a new zenith. The ability to solve problems in the field of computer vision has received a strong thrust with the advent of CNNs. Several CNN approaches have been used for

classification of leaves.

In [8], the authors have classified diseases occurring in plants based on AlexNet model. Nine different plants categorized into two such as diseased and healthy leaves have been selected for the study. The images were resized to the required format. Entire dataset has been divided into two: training and testing data. AlexNet model classified the leaves into two categories and compared performance parameters of accuracy, confusion matrix etc with support vector machine. AlexNet gained an accuracy of 91.15% and SVM with linear kernel, achieved 89.69% accuracy.

The authors of [9] has performed a quality inspection of different sugarcane varieties by using convolutional neural networks. A successful classification increases the yield as well as plant population. Two levels of transfer learning approaches have been used here. In the first level of transfer learning, all models have been pre-trained with a large dataset of a single variety of sugarcane. In the second level of transfer learning, all models are retrained with a small dataset of different varieties of sugarcane billets. Four architectures used for this classification are AlexNet, VGG-16, GoogleNet and ResNet. Last fully connected layer consists of only two neurons, as it contains only two classes. AlexNet attained higher accuracy with lower processing time.

In [10] researchers classified leaves with medicinal values using image processing techniques and artificial neural network. Here six different species of plants from 40 different sites were selected for the experiment. Segmented images from the background are given to the feature extractor for extracting color, texture and shape. Artificial neural network classified the images and mean square error and correct classification rates are evaluated.

In [11] the authors have used MobileNet architecture for detection of apple leaf disease. Since MobileNet architecture is a lightweight deep learning method, it has become easier to deploy on mobile phones. This low cost technique provides higher stability and precision. Training and testing datasets have been selected in the ratio of 3:1. Accuracy and average handling time are the two performance metrics used for the evaluation. MobileNet has been compared with ResNet-152 and InceptionV3. MobileNet achieved higher efficiency. The authors in [12] use transfer learning for detection of leaf disease. Here, ResNet-50 has been used and a comparison was done on its accuracy with other models such as VGG-16, VGG-19 and AlexNet.

In [13] researchers introduce classification techniques for recognition of Malaysian herbs using deep learning algorithms. Self acquired images were used for this work and the coloured images were converted to gray scale image using a pre-processing step. Canny edge detector extracts the edges. Zernike and Hu are the two descriptors that collect shape features. Through gray level co-occurrence matrix technique textural features were extracted. Classifiers such as support vector machine (SVM) with Radial Basis Function kernel and deep learning neural network (DLNN) classify images and mo-

bile app for recognition of herbs have been developed. DLNN achieved higher accuracy when compared with SVM.

In [14], authors created a dataset of forty different species of medicinal herbs found in the southern part of India. The collected images were resized, blurred and given to edge detector for extraction of edges. Salt and pepper noises were removed and grab cut segmentation technique was used to separate the leaf images from the background. Feature extraction plays an ineluctable role in identification of herbs. Four CNN models VGG-16, VGG-19, InceptionV3 and Xception were used. Classifiers such as ANN and SVM categorize the leaf images into different groups. Bayesian optimization was also used along with SVM. Xception along with ANN attained higher accuracy of 97.5%. A mobile application developed with this model was found to be capable of predicting herbs within a matter of a few seconds, thus taking another step in the inexorable march towards automation in all fields. Hardware architectures for CNNs are also studied in recent literature [15].

3 Proposed Approach

CNNs are the backbone of image classification tasks and provide dramatic improvements in performance compared to traditional image processing techniques. The advancements in CNN generate more complex and accurate computer vision models with automatic feature extraction.

3.1 Dataset

This study has been done using a dataset from freely accessible Mendeley medicinal leaf dataset. 1835 images comprising 60 to 100 segmented, colored and high quality images of 30 different species of medicinal herbs like Sandalwood, Mint, Mustard, Jamica cherry etc are used. In order to make the dataset available to deep learning techniques, data augmentation was done. The data is split into two categories: one set for training and the other for testing purposes. A ratio of 80:20 is used for training and testing respectively. Evaluation of the model was done based on the test data.

3.2 CNN Models

In this section we discuss various CNN models that are being used in this work for classification of leaves.

3.2.1 VGGNet

VGGNet is a convolutional neural network proposed by A Zisserman and K Simonyan [16]. The architecture comprises of convolutional layers, max pooling and fully connected layers. Convolutional layers of 3×3 kernels are followed by ReLU. ReLU, an activation function will produce an output when the input is positive or it results in zero. Max pooling reduces dimensionality. Fully connected layers and softmax layers are added as final layers [17].

3.2.2 ResNet

Residual Network (ResNet) is a convolutional neural network that uses the concept of skip connection [18]. Vanishing gradient problem is resolved in skip connection by skipping layers. ResNet is eight times deeper than VGGNets. It consists of convolutional layers, pooling layers and fully connected layers. Batch normalization is introduced between convolutional layer and activation function. Two types of skip connections are used in residual networks and it depends upon the dimensions of both input and output. If the input and output activation are in the same dimension, then identity blocks are used. On the other hand, if it is not in the same dimension, convolutional blocks need to be inserted. Convolutional block uses CONV2D layer in the shortcut path.

3.2.3 MobileNet

MobileNet is a lightweight convolutional neural network. The main aim of MobileNet architecture is to make the size of the system small and maximize speed [19]. Depthwise separable convolution used in MobileNet reduces the number of parameters. Two operations involved in this architecture are depthwise convolution followed by pointwise convolution. Depthwise convolution applies a single filter to each input channel. Pointwise convolution applies 1×1 convolution to the output of depthwise convolution. MobileNetV2 is the second version of MobileNet. Three convolution layers were used in this architecture along with residual connection. First convolution block is the expansion layer as it expands the number of channels before giving it to depthwise convolution. The second block is same as that of MobileNet. Third block acts as a bottleneck layer that reduces data flow. Residual network removes the problem of vanishing gradient.

3.2.4 DenseNet

DenseNet is a deep convolutional neural network in which all layers are interconnected [20]. Each layer receives feature maps from all previous layers, thus reducing vanishing gradient problem. Dense blocks and transition blocks are the main building blocks of DenseNet. Each dense layer consists of batch normalization, ReLU, 3×3 convolution and drop out [21]. Transition blocks consist of batch normalization, ReLU, 1×1 convolution, drop out and pooling layer. Size of the feature map should remain the same when we do concatenation and so this operation is performed in the transition block. Connection between layers enhances feature propagation and concatenation encourages feature reuse.

3.2.5 EfficientNet

EfficientNet is a convolutional neural network that performs scaling on depth, width and resolution. Deeper networks have the ability to capture more complex features from images but it will have the disadvantage of higher training time due to vanishing gradient. Width scaling works well only for small models. The finer features can easily be captured from high resolution images but it will result in diminishing accuracy gain. Hence balancing of all dimension scaling is very important for achieving better efficiency and accuracy. Compound scaling is an efficient remedy for this issue. Constant

coefficients are determined by a small grid search technique. EfficientNetB0 is the baseline model. Mobile inverted bottleneck convolutional layers and squeeze and excite blocks are the main building blocks of EfficientNet architecture [22]. EfficientNetV2M is the second version of EfficientNet model and provides higher training speeds and a better parameter efficiency. It introduces training aware architecture search and it chooses the best combination of MBConv, MBFusedConv, number of layers, kernel size and expansion ratio. Progressive learning with adaptive regularization is the other contribution of EfficientNetV2 model and it outperforms vision transformer [23].

4 Implementation and Results

4.1 Experiments

Deep learning models are developing very fast and image classification tasks with smaller datasets are taking advantage of pre-trained models. Knowledge gained by convolutional neural network from previous experience can be transferred to other high similarity, low quantity cases resulting in higher efficiency and lesser training time. The block diagram of pre-trained models are shown in Figure 1.

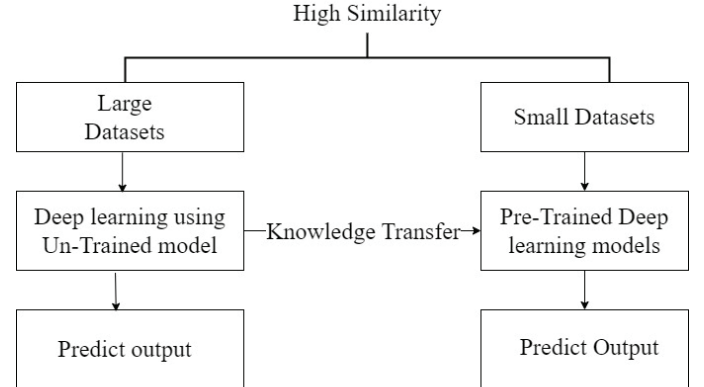


Figure 1: Block diagram of pre-trained models

In Deep learning models, initial layers are used predominantly for feature extraction and the final layers for image classification. VGG-16, ResNet-50, DenseNet-121, MobileNetV2, EfficientNetB0 and EfficientV2M are some of the CNN models taken for this study. These models have been pretrained to the imagenet dataset comprising of 12 lakh images under 1000 different categories. The knowledge acquired from this training has been used to train new datasets. The time required for training fine-tuned models is relatively much less when compared with that of models built from scratch. New datasets are taken from Mendley medicinal leaf dataset which comprises of 1835 images belonging to 30 different species. Images are used for training and validation purposes in a ratio of 80:20 adding up to 1468 and 367 images respectively. The images have been resized in order to be compatible with the deep learning models. The experiments were performed on graphics processing unit (GPU) platform. The training of images is carried out in 32 batches and uses adam for optimization. Initial layers of all models were kept unaltered and a dense

layer of 30 with softmax layer is added to final layers. Each of the experiments are carried out in 5 and 15 epochs. The performance metrics such as accuracy and loss are monitored and noted.

4.2 Results and Discussion

In this study, an analysis of six CNN models have been done for the task of leaf classification. The performance of various models are shown in Fig.2-7. Accuracy and loss per epoch of each of the models have been plotted in graphs.

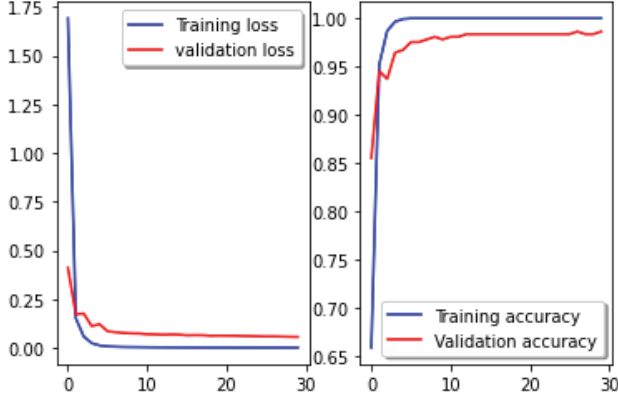


Figure 2: Accuracy and loss per epoch of VGG-16

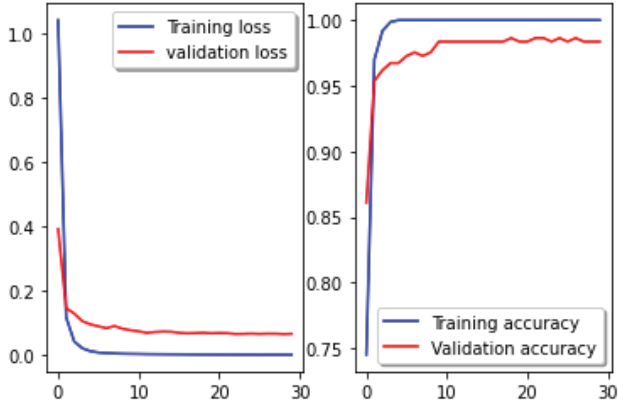


Figure 3: Accuracy and loss per epoch of ResNet-50

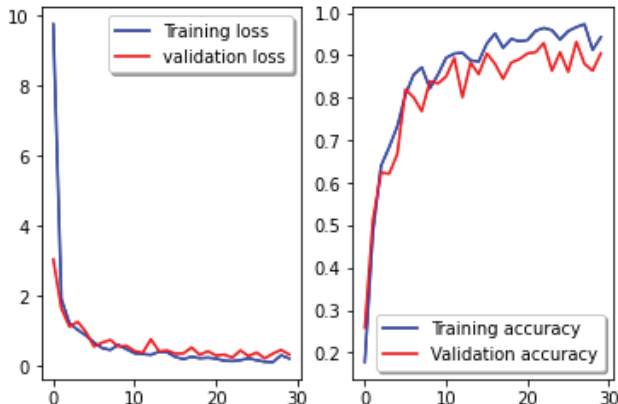


Figure 4: Accuracy and loss per epoch of DenseNet-121

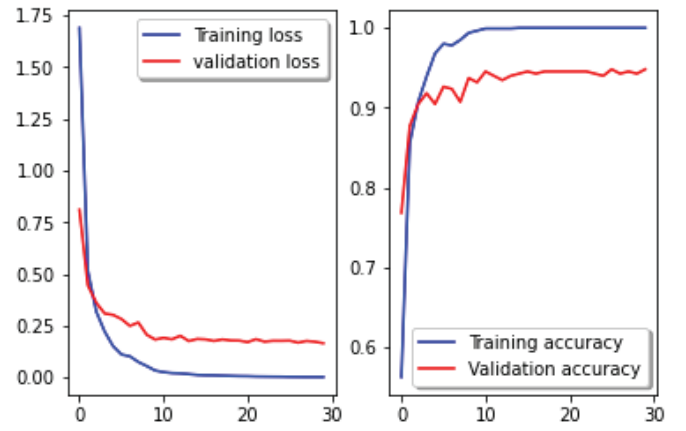


Figure 5: Accuracy and loss per epoch of MobileNetV2

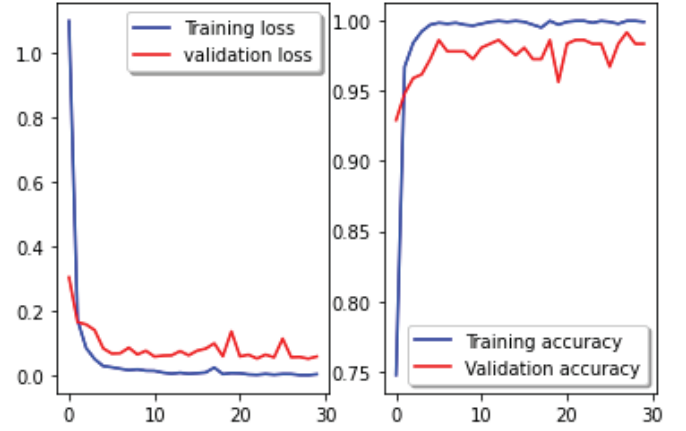


Figure 6: Accuracy and model loss per epoch of EfficientNetB0

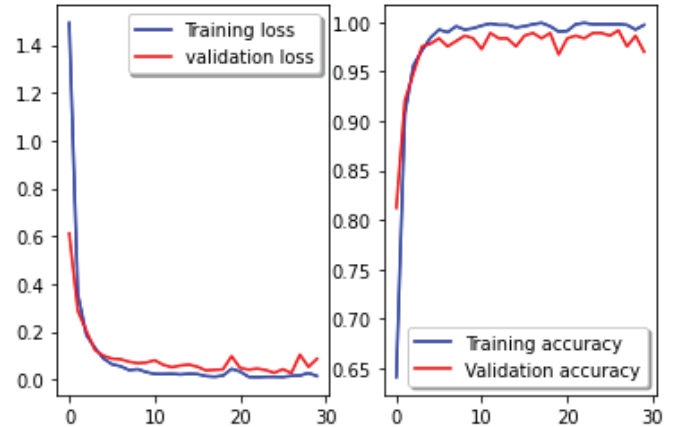


Figure 7: Accuracy and loss per epoch of EfficientNetV2M

Table 1 analyzes performance of all models for 5 epochs. All architectures except DenseNet attained accuracy above 80%. EfficientV2M and EfficientB0 achieved the highest results with an accuracy level of 97.5%. Accuracies and losses according to epochs are described in the table. It has been noted that DenseNet performs poorly with fewer iterations.

Table 1: Comparison of CNNs for leaf classification

Model	5 epochs				15 epochs			
	Training accuracy (%)	Test accuracy (%)	Training loss	Test loss	Training accuracy (%)	Test accuracy (%)	Training loss	Test loss
VGG-16	100	96.7	0.0059	0.0905	100	98.6	0.0011	0.0553
ResNet-50	100	97.5	0.0095	0.0907	100	98.9	0.0014	0.0663
DenseNet-121	91.9	86.9	0.2427	0.5034	93.3	86.9	0.2267	0.4204
MobileNetV2	97.4	90.4	0.1282	0.3119	100	93.7	0.0139	0.1906
EfficientNetB0	99.9	97.5	0.0172	0.1088	100	97.8	0.0027	0.0675
EfficientNetV2M	99.7	97.5	0.0364	0.1120	100	98.3	0.0044	0.0624

Table 1 also compared different models for 15 epochs. After 15 epochs ResNet-50, VGG-16 and EfficientV2M provides maximum accuracy of 98%. Among the three models EfficientV2M converges faster and hence EfficientV2M gives best results with fewer number of epochs.

5 Conclusion

Classification of herbs into different categories is an essential field in ayurvedic research. Convolutional neural networks perform well for leaf classification. In this work, fine-tuning and evaluation of the most advanced deep learning models are performed. The architectures included in this study are VGG-16, ResNet-50, DenseNet-121, MobileNetV2, EfficientB0 and EfficientV2M. EfficientNetV2M achieved higher accuracy rates of 98% with fewer number of epochs. Thus deep learning networks can easily replace human interventions in the recognition of herbs. As a future scope of this work, handheld devices based on CNN can be developed to automate the process of herb recognition.

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